Joint inversion
Multigeophysical inversion

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Ketil Hokstad, April 2021
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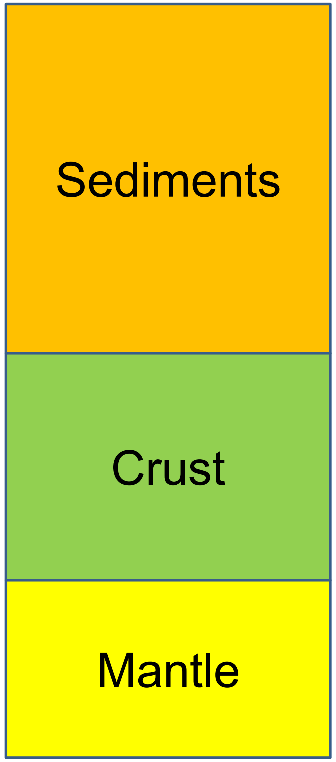
Joint inversion and multigeophysical inversion

Two main types of objectives

- Structural imaging
- Estimating properties of interest

Many scales and applications

- Petroleum exploration
  - Structural imaging
  - Fluid prediction
  - Basin modeling
- Geothermal exploration
- Mineral exploration
- Academic applications



- AVO and CSEM inversion
  - ➢ Prospect risking
- PSDM: Tomography/FWI, MT, FTG inversion
  - ➢ Improved imaging
- Crustal properties: Gravity, magnetic, MT, refractions
  - ➢ COB, RHP, heat flow
  - ➢ Hydrothermal systems
- Mantle properties: Gravity, topography, geoid
  - ➢ Depth to LAB => heat flow

From joint interpretation to joint inversion

We have always done joint interpretation in exploration

Petroleum

«Petroleum system analysis»

- Source
- Reservoir
- Trap/Seal

Geothermal

«Play fairway analysis»

- Heat source
- Recharge
- Producibility

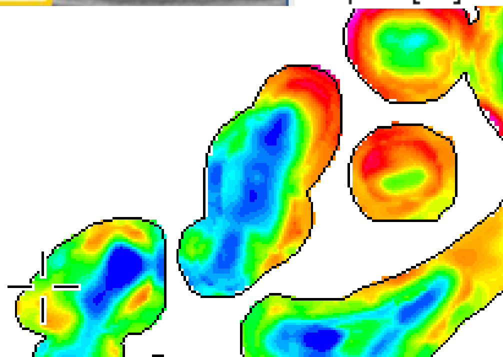
Seabed massive sulfides

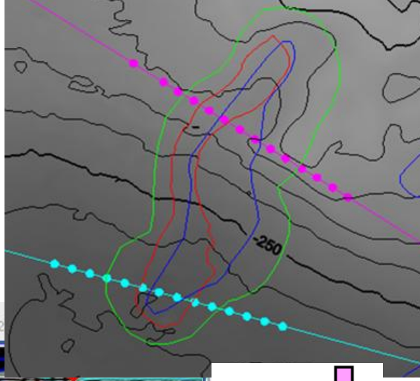
«Mineral systems analysis»

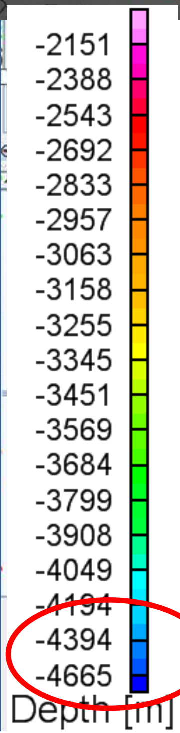
- Metal source
- Fluid pathway
- Accumulation and preservation

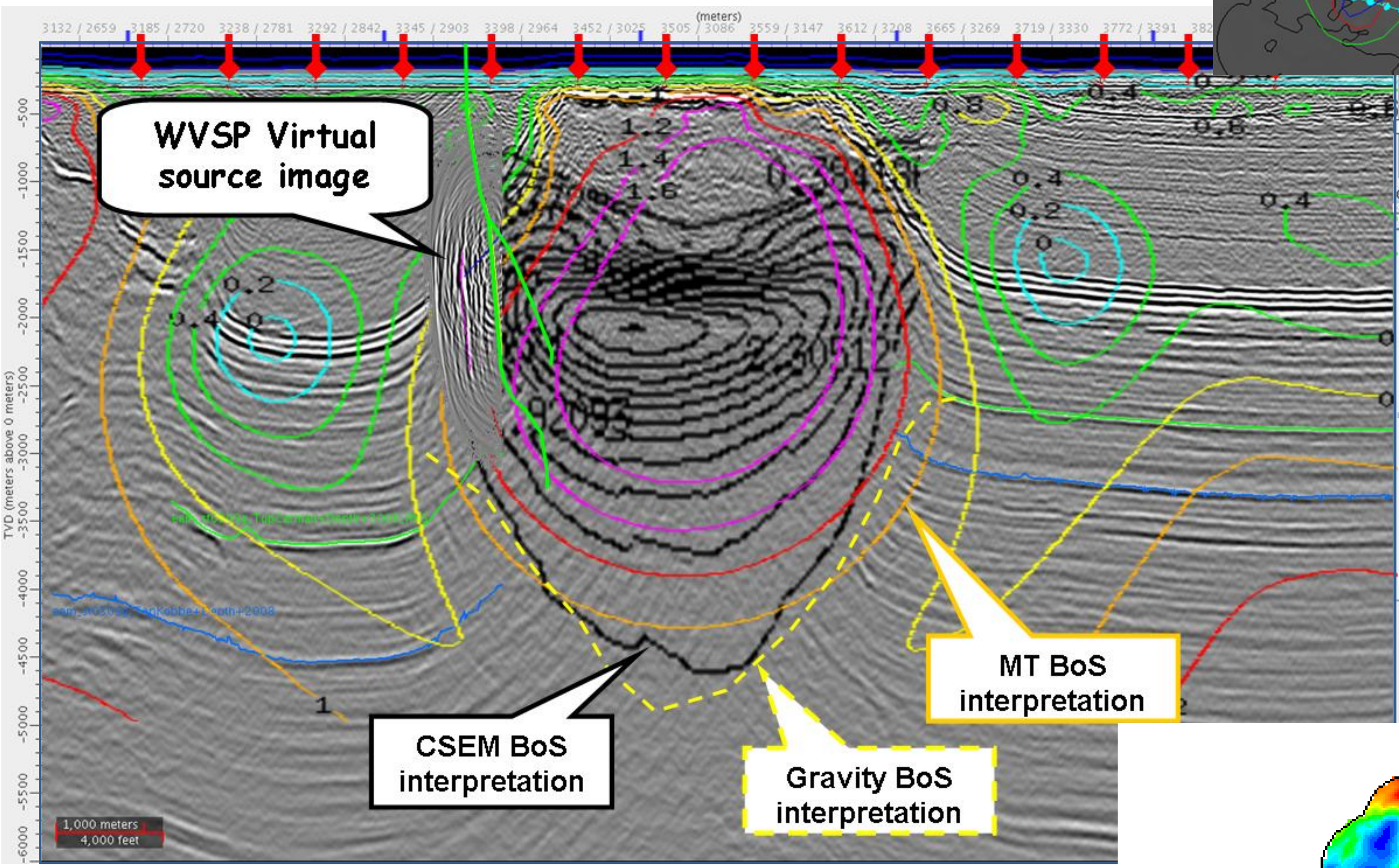
If one play element fails, the play fails

From joint interpretation to joint inversion
Uranus well line – joint interpretation



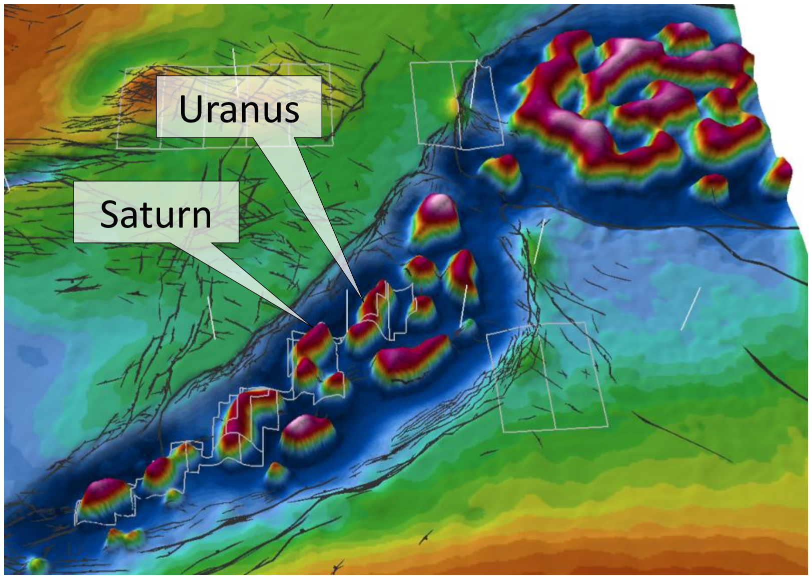


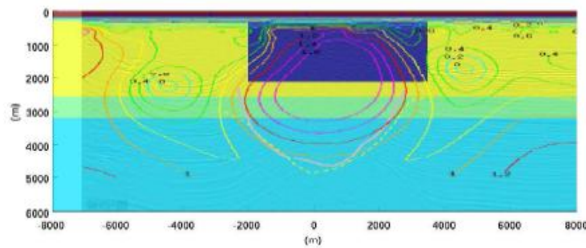




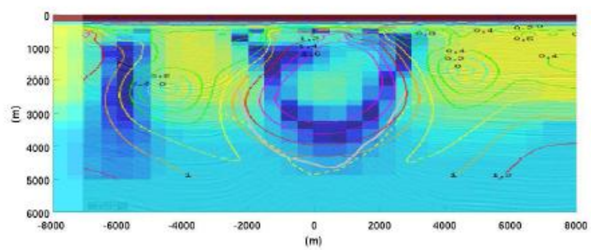
Hokstad et al. (2011), Joint Imaging of Geophysical Data:
Case History from the Nordkapp Basin, Barents Sea

Contrast source inversion (Wiik et al, 2013): Nordkapp Basin joint CSEM and MT inversion

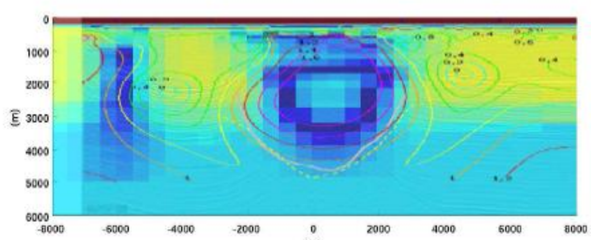




Initial model



MT inversion
(E and H fields)



MT+CSEM inversion

Dominating contributions;

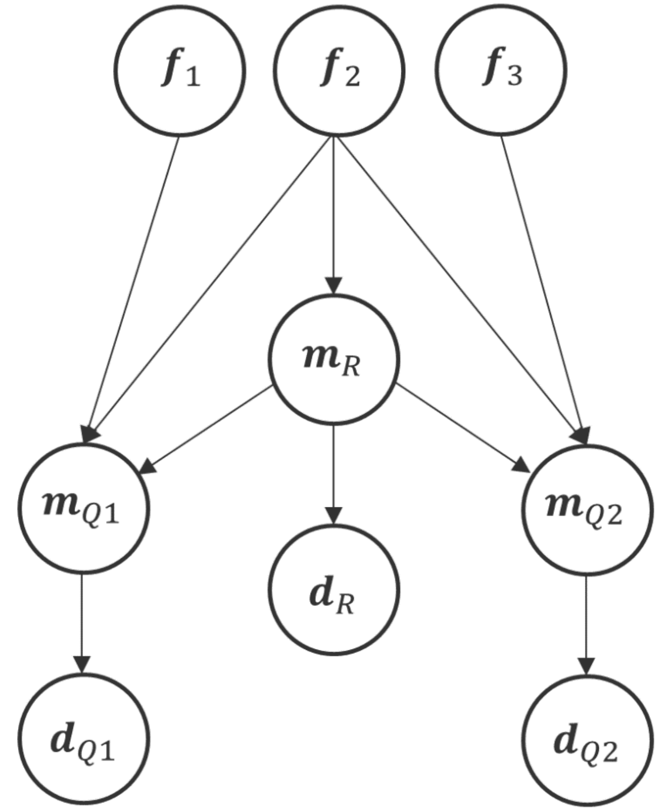
- CSEM => vertical resistivity
- MT => horizontal resistivity

Generic SJI/MGI formalism

Two different approaches

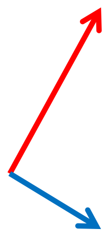
- Statistical
  - Bayesian inversion
  - Including posterior uncertainty
- Optimization
  - BFGS, L-BFGS
  - Gauss Newton …
- Both can be derived from a common starting point

- **Generic formulation for any multi-geophysical inverse problem**



Bayesian network (DAG) where model $\boldsymbol{m}_R$ acts as a structural reference for models $\boldsymbol{m}_{Qi}$

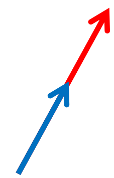
Simultanouos Joint Inversion (SJI)
Cross-gradient principle



Orthogonal vectors:

$$\mathbf{a} \cdot \mathbf{b} = 0$$

$$a_i b_i = 0$$



Parallel vectors:

$$\mathbf{a} \times \mathbf{b} = (0,0,0)$$

$$\epsilon_{ijk} a_j b_k = 0$$

<u>Cartesian</u> model gradient

$$\mathbf{h}_{\mathrm{Q}} = \nabla \mathbf{m}_{\mathrm{Q}}(\mathbf{x})$$

Cross-gradient objective function

$$\Phi_{\mathrm{XG}} = \left[\mathbf{h}_{\mathrm{R}} \times \mathbf{h}_{\mathrm{Q}}\right]^{\mathrm{T}} \boldsymbol{\Sigma}_{\mathrm{XG}}^{-1} \left[\mathbf{h}_{\mathrm{R}} \times \mathbf{h}_{\mathrm{Q}}\right]$$

$$p(\mathbf{m}_Q | \mathbf{m}_R) \quad \propto \quad \mathrm{e}^{-\Phi_{XG}}$$

X-gradient
objective function

Promote structural similarity of different geophysial models

SJI objective function – structural constraints

$$\Phi_{SJI} = \Phi_{MLH} + \Phi_{pri} + \Phi_{sm} + \Phi_{RP} + \Phi_{XG}$$

Single-domain inversions:
- Data fit (MLH)
- Model prior
- Smoothness

SJI terms:
- Rock physics models (Gardner, mudrock)
- Cross-gradients

$$\Phi_{XG} \quad = \quad \frac{1}{2} \sum_{Q \neq R} [\nabla \mathbf{m}_R \times \nabla \mathbf{m}_Q]^T \mathbf{\Sigma}_{XGQ}^{-1} [\nabla \mathbf{m}_R \times \nabla \mathbf{m}_Q]$$ X-gradient

$$\Phi_{RP} \quad = \quad \frac{1}{2} \sum_{Q} [\mathbf{m}_Q - \mathcal{G}_Q(\mathbf{m}_R)]^T \mathbf{\Sigma}_{RPQ}^{-1} [\mathbf{m}_Q - \mathcal{G}_Q(\mathbf{m}_R)]$$ Rock physics

Optimization approach – least squares

- Minimization of total SJI objective function Φ_{SJI}

$$\Delta\mathbf{m} = -\mathbf{H}^{-1}\mathbf{g}$$

$$\mathbf{g} = \frac{\partial\Phi}{\partial\mathbf{m}}$$ gradient

$$\mathbf{H} = \frac{\partial^2\Phi}{\partial\mathbf{m}^2}$$ Hessian

- Written out:

$$\Delta\mathbf{m}_Q = -[\mathbf{H}_Q^{MLH} + \mathbf{H}_Q^{pri} + \mathbf{H}_Q^{sm} + \mathbf{H}_Q^{RP} + \mathbf{H}_Q^{XG}]^{-1}[\mathbf{g}_Q^{MLH} + \mathbf{g}_Q^{pri} + \mathbf{g}_Q^{sm} + \mathbf{g}_Q^{RP} + \mathbf{g}_Q^{XG}]$$

- Need to broadcast gradients and Hessians from single-domain inversions
- Finds the (local) minimum of the total SJI objective function

Simplified SJI approach

- Simplified split-step approach:

  1. Single domain inversion

$$\Delta \tilde{\mathbf{m}}_Q \quad = \quad -[\mathbf{H}_Q^{MLH} + \mathbf{H}_Q^{pri} + \mathbf{H}_Q^{sm}]^{-1}[\mathbf{g}_Q^{MLH} + \mathbf{g}_Q^{pri} + \mathbf{g}_Q^{sm}]$$

  2. SJI update

$$\Delta \mathbf{m}_Q \quad = \quad \Delta \tilde{\mathbf{m}}_Q - [\mathbf{H}_Q^{RP} + \mathbf{H}_Q^{XG}]^{-1}[\mathbf{g}_Q^{RP} + \mathbf{g}_Q^{XG}]$$

- Does not need to broadcast single domain-gradients and Hessians
- Does generally not find the (local) minimum of the full SJI objective function

X-gradient terms - details

Model-parameter gradient:

$$\mathbf{g}_Q^{XG} \quad = \quad [\mathbf{h_R} \times \mathbf{h_Q}]^\mathbf{T} \boldsymbol{\Sigma}_\mathbf{XG}^{-1} [\mathbf{h_R} \times \frac{\partial \mathbf{h_Q}}{\partial \mathbf{m_Q}}]$$

Hessian (Gauss-Newton):

$$\mathbf{H}_Q^{XG} \quad = \quad \left[\mathbf{h_R} \times \frac{\partial \mathbf{h_Q}}{\partial \mathbf{m_Q}}\right]^T \boldsymbol{\Sigma}_{XG}^{-1} \left[\mathbf{h_R} \times \frac{\partial \mathbf{h_Q}}{\partial \mathbf{m_Q}}\right]$$

Cartesian gradient and derivatives:

$$\mathbf{h_Q} = \nabla \mathbf{m_Q}$$

$$\frac{\partial h_{Qki}}{\partial m_{Qk}} = \sum_{j=1}^{3} \frac{\partial h_{Qki}}{\partial x_j} \frac{\partial x_j}{\partial m_{Qk}} = \sum_{j=1}^{3} \frac{1}{h_{QKj}} \frac{\partial h_{Qki}}{\partial x_j}$$

$$\frac{\partial h_{Qki}}{\partial m_{Qk}} = \frac{\partial}{\partial x_i} \ln \left(\prod_{j=1}^{3} h_{Qkj} \right)$$

$Q \in \{\rho, v_P, v_S, R, M, \cdots\}$

$k \in \{1,2,\cdots, N_{Qm}\}$

$i, j \in \{1,2,3\}$

Published work on SJI

Moorkamp et al (GJI, 2011) and di Stefano et al (Geophysics, 2011) use a similar X-gradient objective function as us:

$$\Phi_{XG} = \frac{1}{2}\sum_{Q\neq R}[\nabla\mathbf{m}_R \times \nabla\mathbf{m}_Q]^T\mathbf{\Sigma}_{XGQ}^{-1}[\nabla\mathbf{m}_R \times \nabla\mathbf{m}_Q]$$

X-gradient

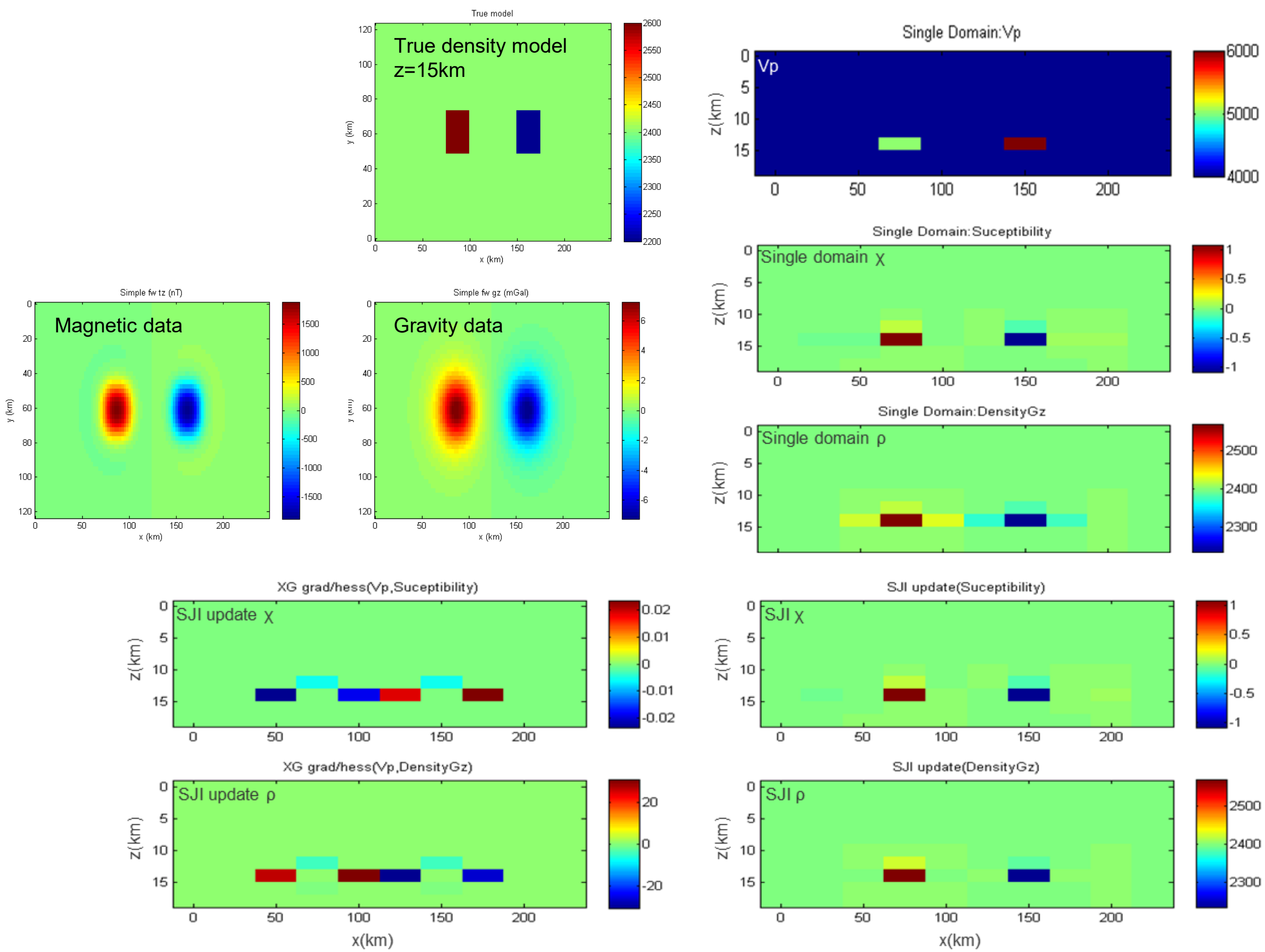
Gallardo (GRL, 2007) use a formulation based on Lagrange multipliers:

$$\min\left\{\sum_{i=1}\phi_{di} + \sum_{j=1}\phi_{Lj} + \sum_{p=1}\phi_{0p}\right\}$$

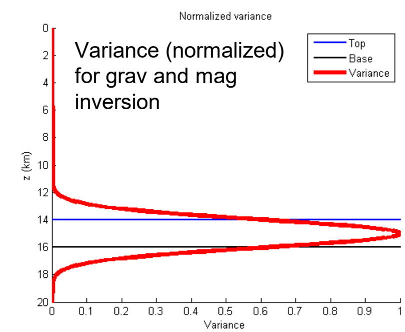
$$\textit{subject to } \tau_g = 0.$$

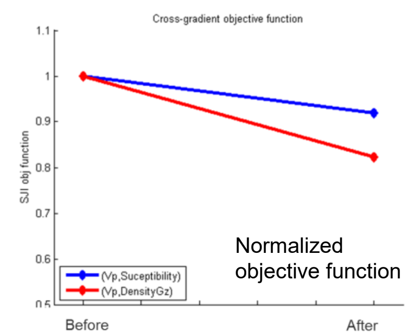
X-gradient

Synthetic example: Joint grav-mag inversion



Velocity model: Structural constraint (not updated in the joint inversion)





Nordkapp Basin example: Cross-gradient

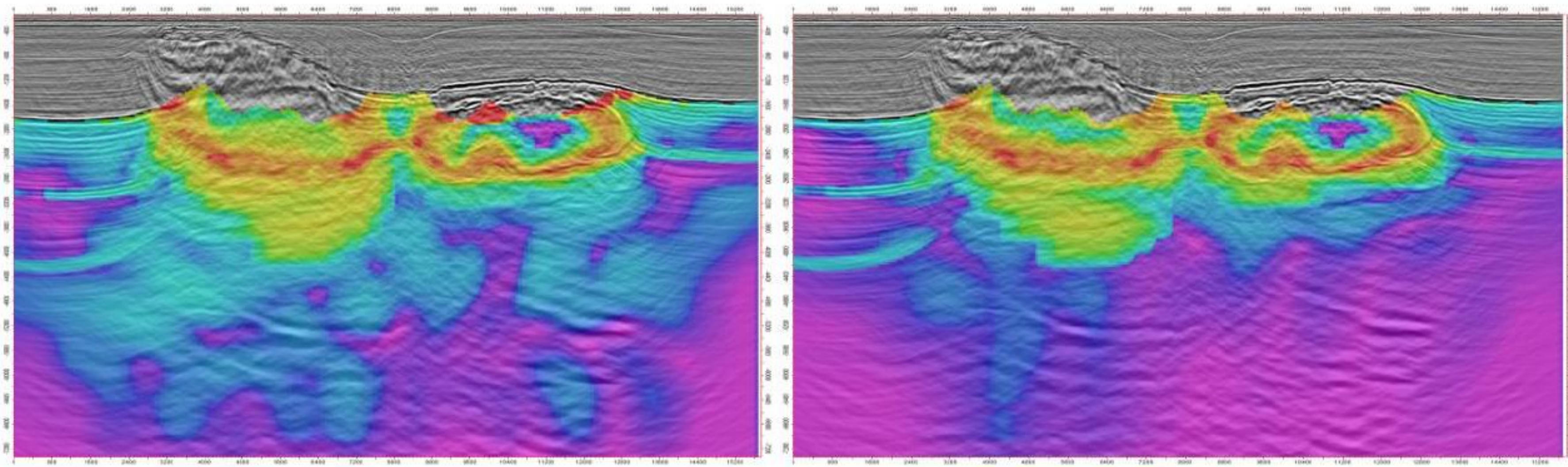


Figure 6 *Cross-Gradient functional display between single domain seismic and gravity inversion results (left), and between SJI velocity and density property (right). The overall decrease of Cross-Gradient indicates a more adherent solution to structural concordance hypothesis for SJI.*

Nordkapp Basin example: PSDM vs SJI

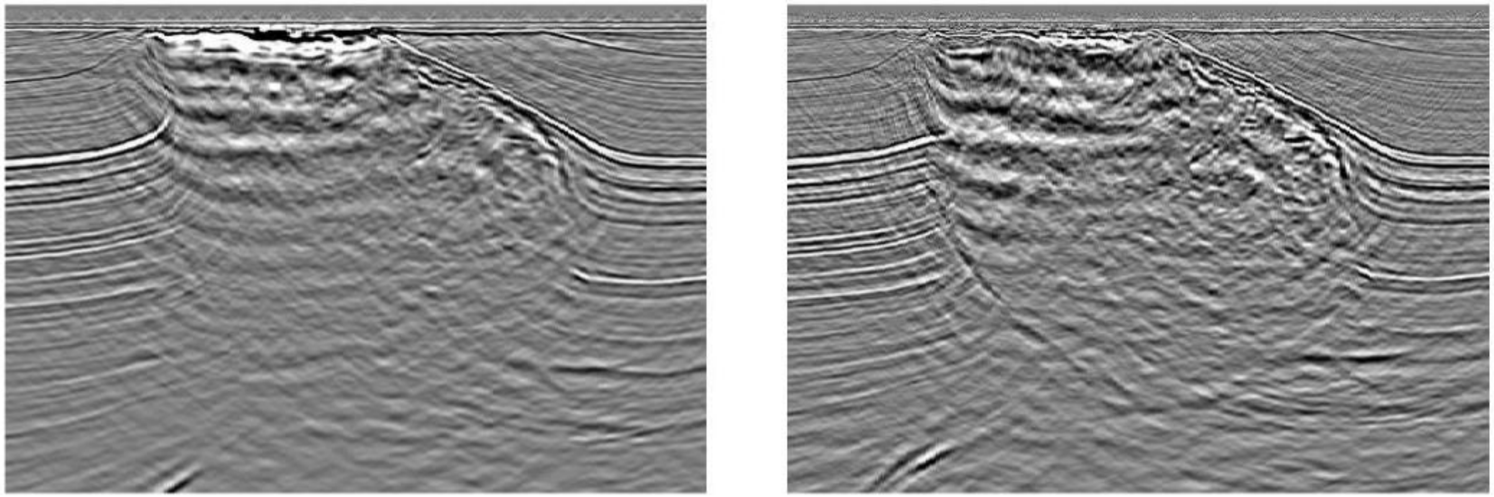
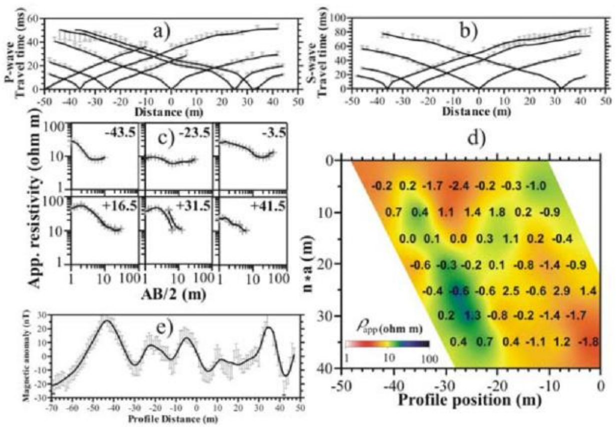


Figure 5 *Seismic tomography image before SJI (left) and SJI image after grid based Cross Gradient update (right).*

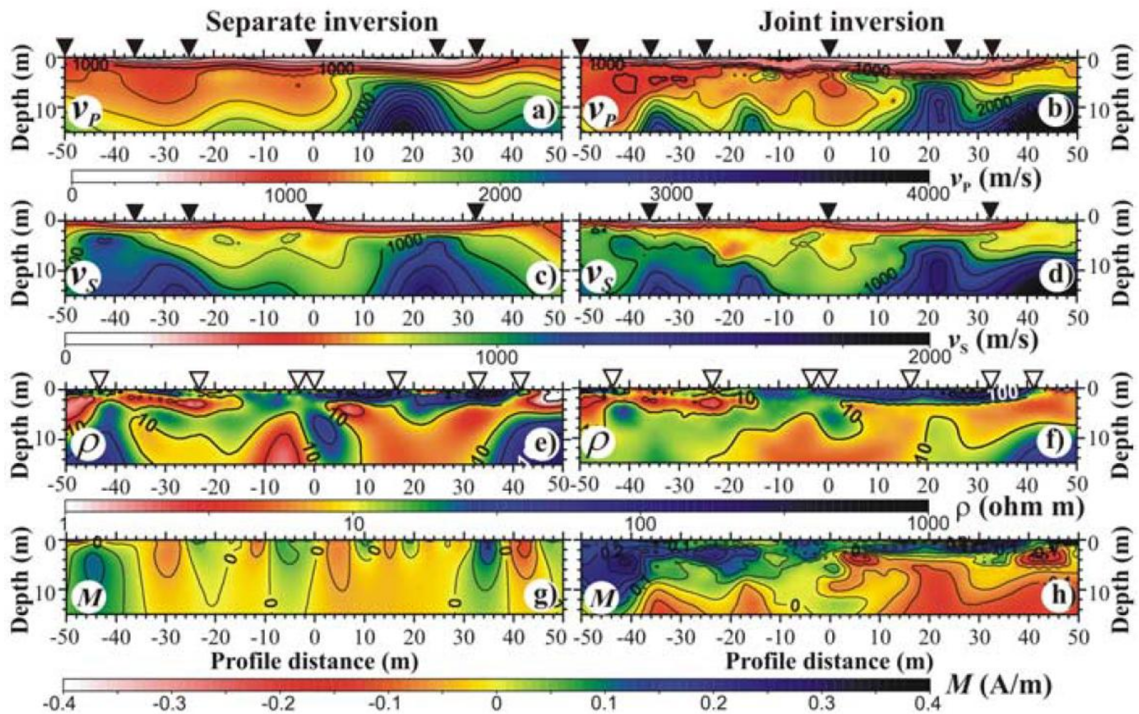
From Mantovani et al, EAGE (2014) and diStefano et al (2011)

SJI example from Gallardo (GRL, 2007)



Data:

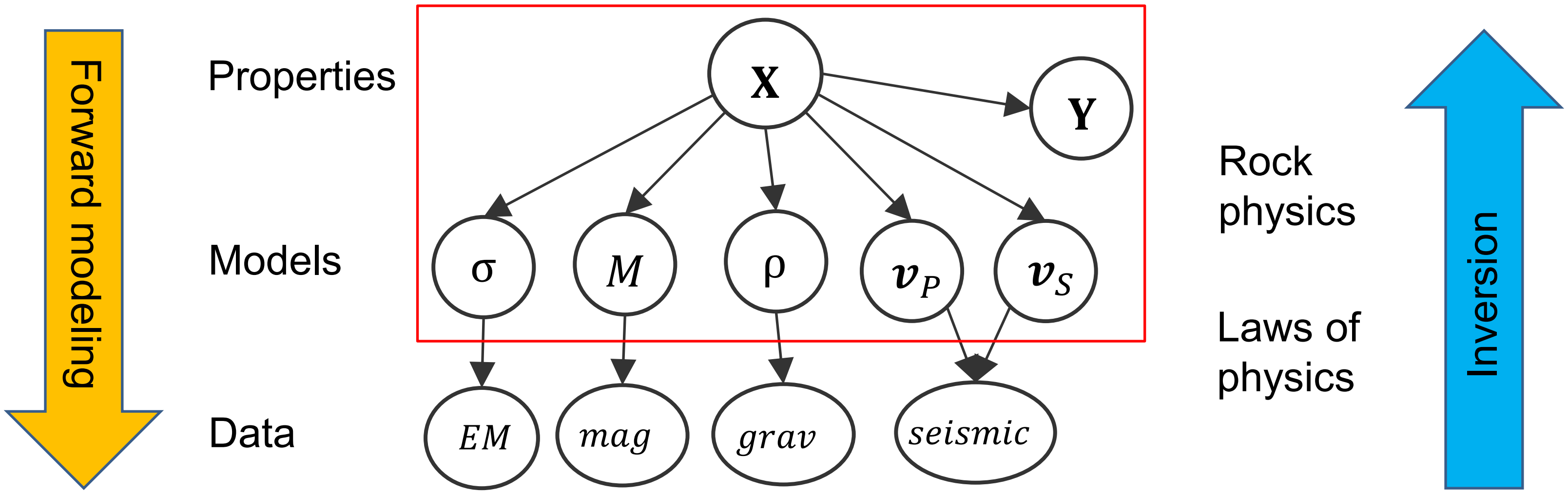
a) P-wave traveltimes
b) S-wave traveltimes
c) Vertical DC electric
d) Dipole-dipole EM
e) Magnetic data



Single domain inversions

Joint inversion

Multigeophysical inversion
Estimating properties of interest

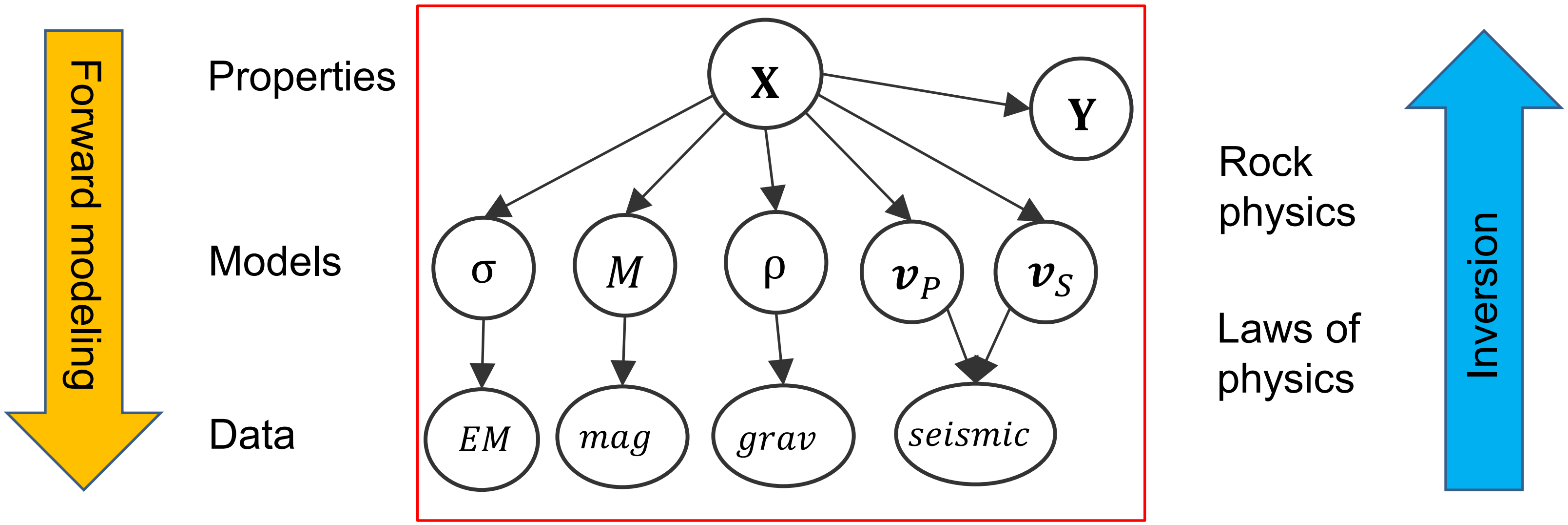


Bayesian network (DAG)

$$p(\mathbf{X}|m_1,\cdots,m_n) = C\prod_{i=1}^{n} p(m_i|\mathbf{X})p(\mathbf{X};\lambda)$$

- Conditional independence
- Independence of noise

Multigeophysical inversion
Estimating properties of interest

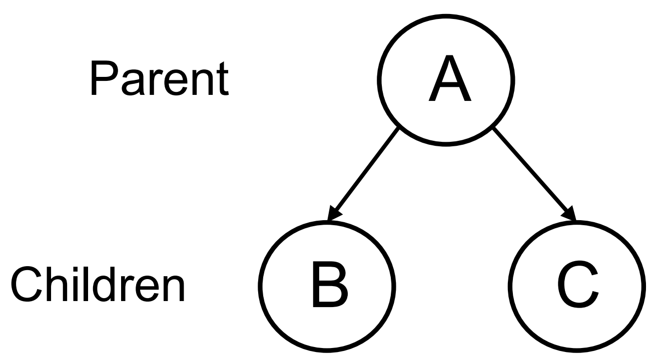


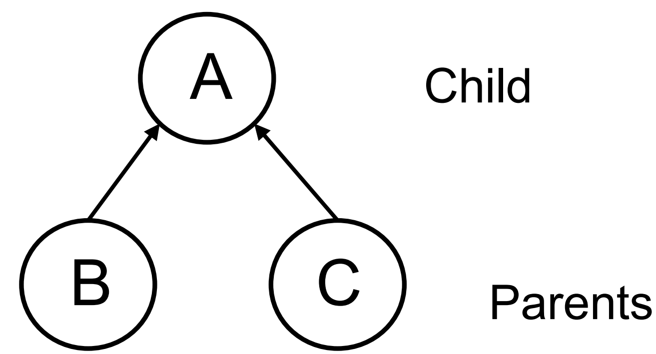
Complete expression; marginalization

$$p(\mathbf{X}|\mathbf{d}) = \int p(\mathbf{X}|\mathbf{m})p(\mathbf{m}|\mathbf{d})d\mathbf{m}$$

Bayesian networks – the general formula

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} p(x_i | x_i^{pa})$$





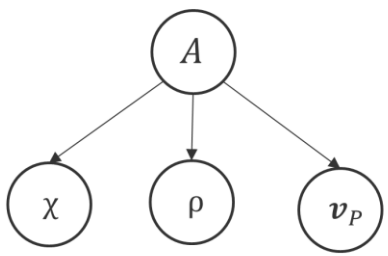
$p(A, B, C) = p(B|A)p(C|A)P(A)$

$p(A, B, C) = p(A|B, C)P(B, C)$

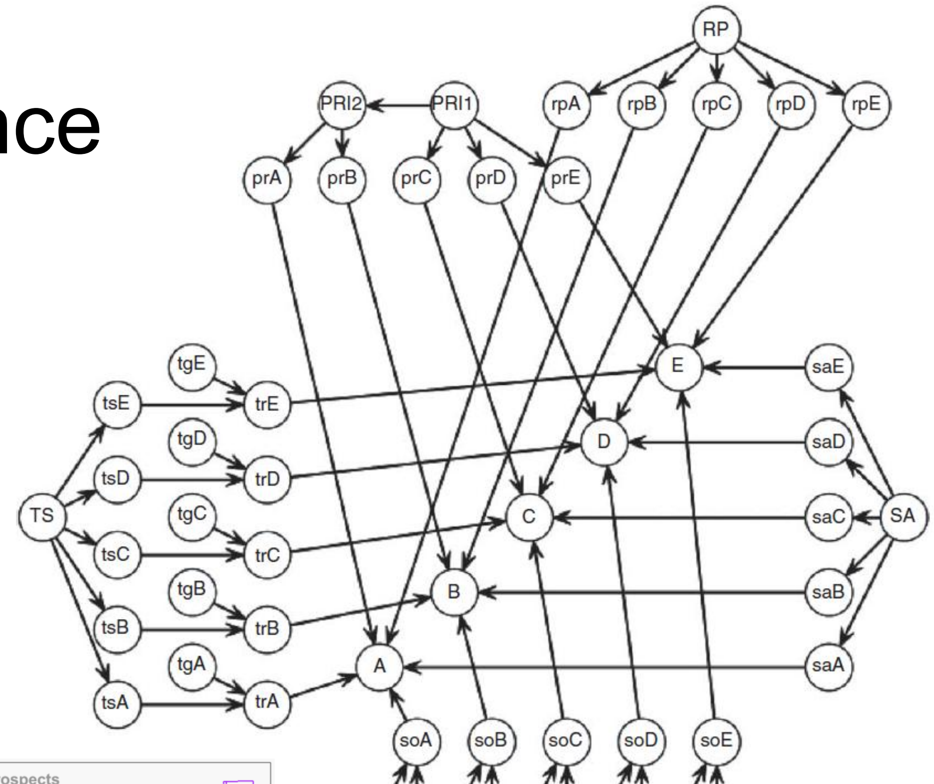
Bayesian networks; conditional independence

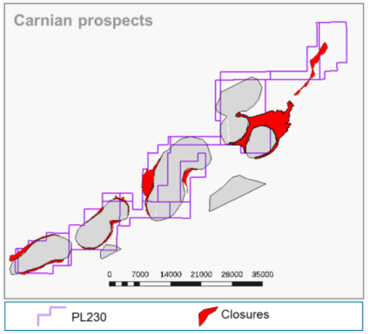
Joint probability factorization

$$p(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} p(x_i | x_i^{pa})$$



Simple Bayesian network (DAG) for rock physics inversion

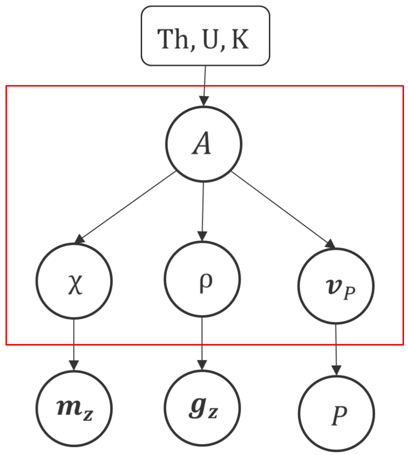


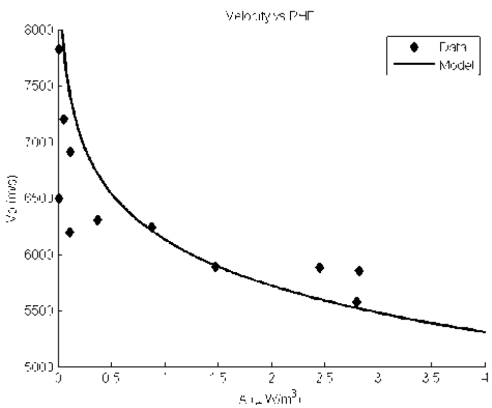


(Martinelli et al., SPE, 2014)

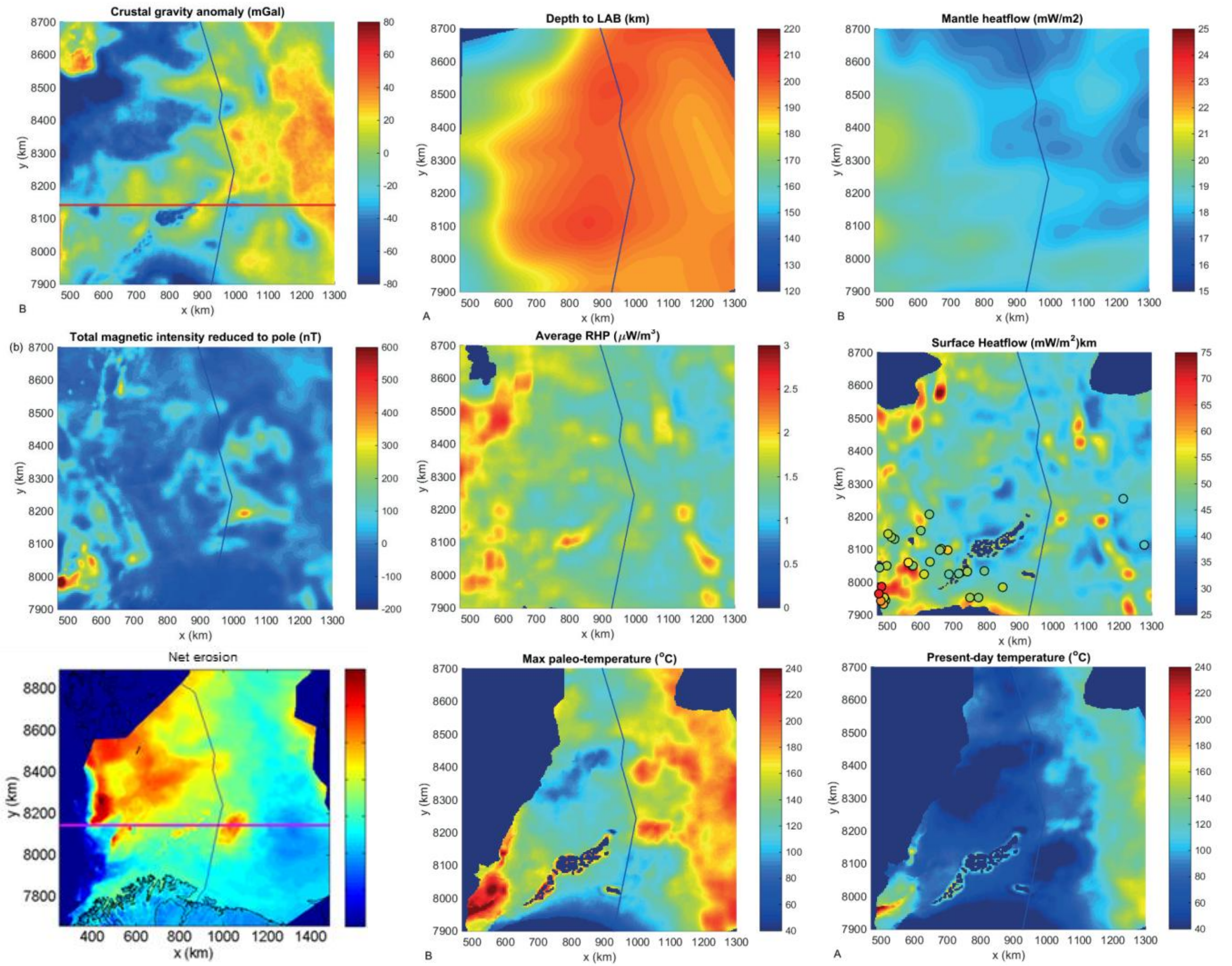
Fig. 3—BN representation of the basin. The Prospects A through E are in the center nodes. Their immediate parents are the observable success factors, and these are again linked through networks. On the left is the trap subnetwork, on the top the producibility (PR) and the reservoir presence (RP) subnetworks, on the right the source abundance (SA) subnetwork, on the bottom the source subnetwork.

Petroleum exploration: RHP and heat flow



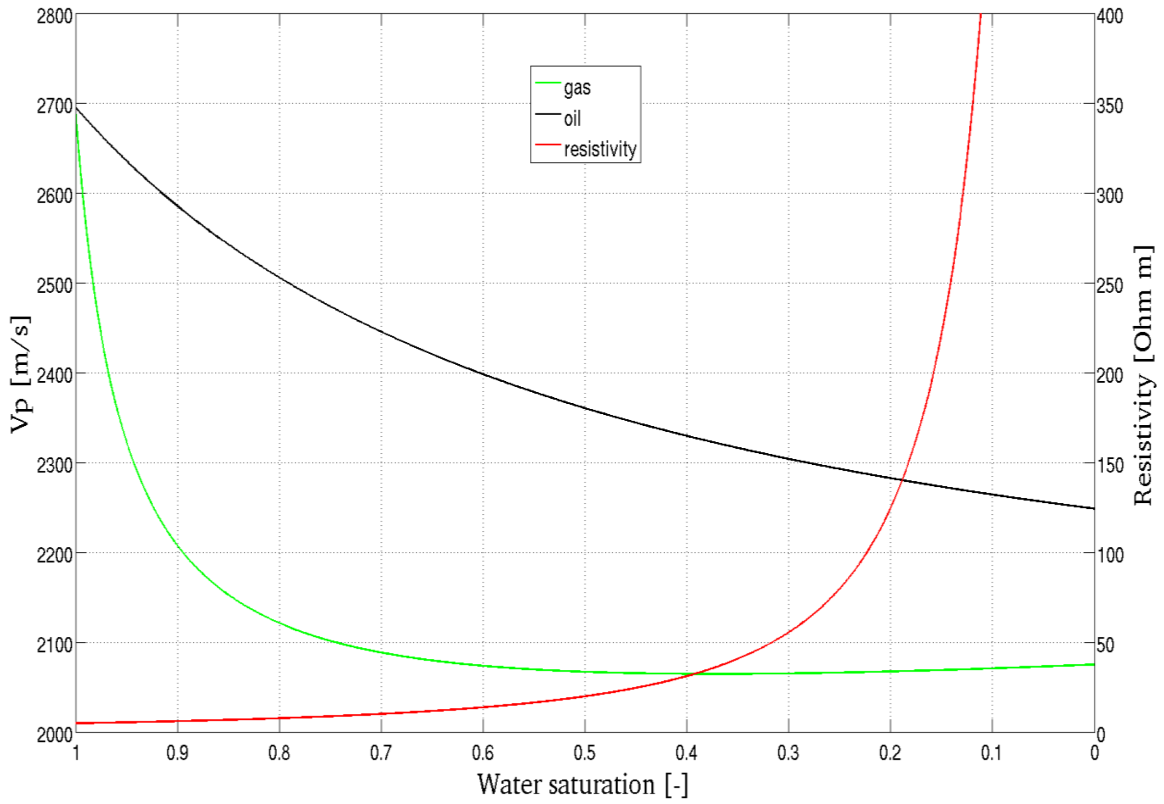


Crustal rock physics relations



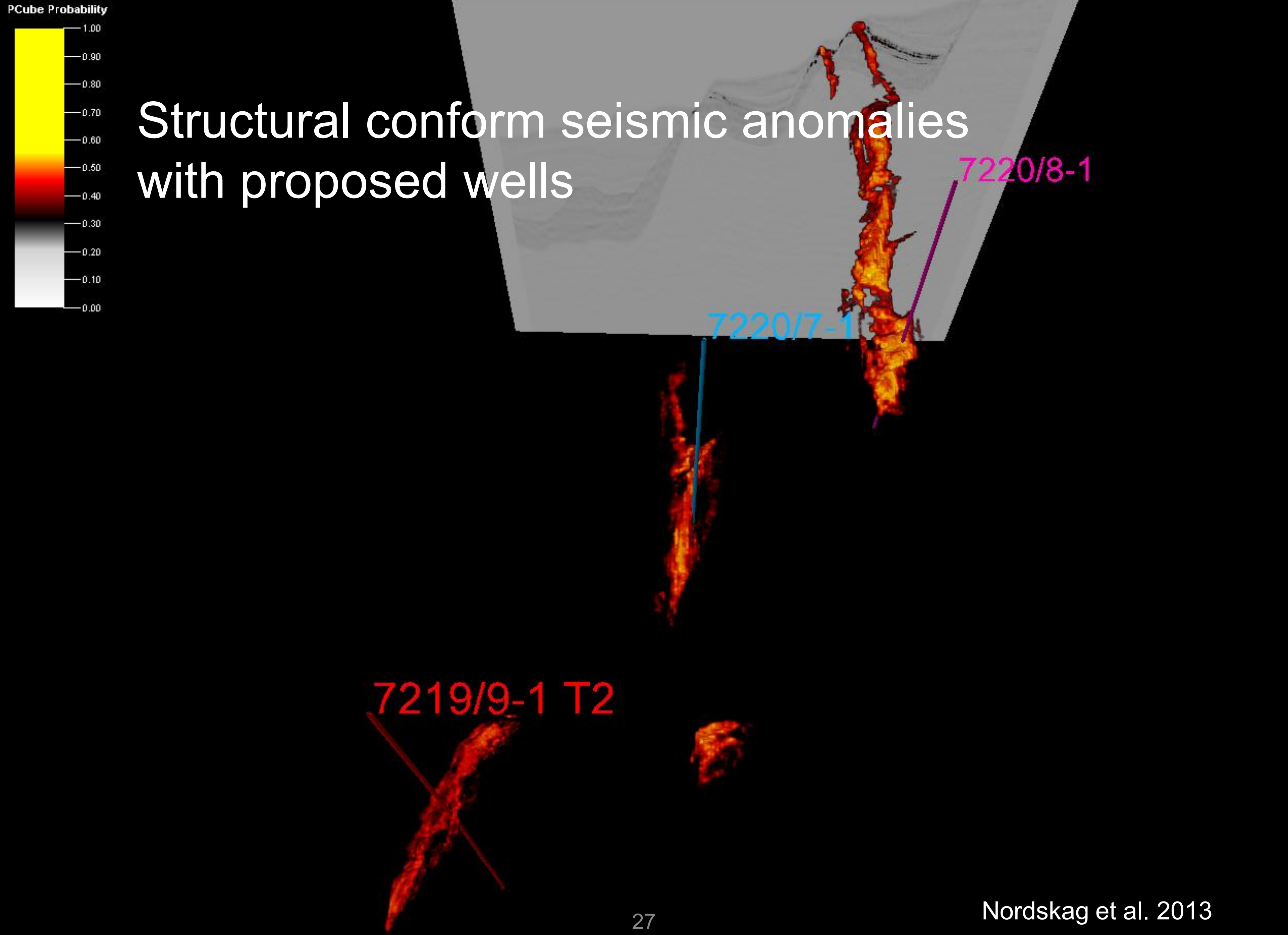
Hokstad et al. (NJG, 2017)

Fluid prediction from seismic and CSEM



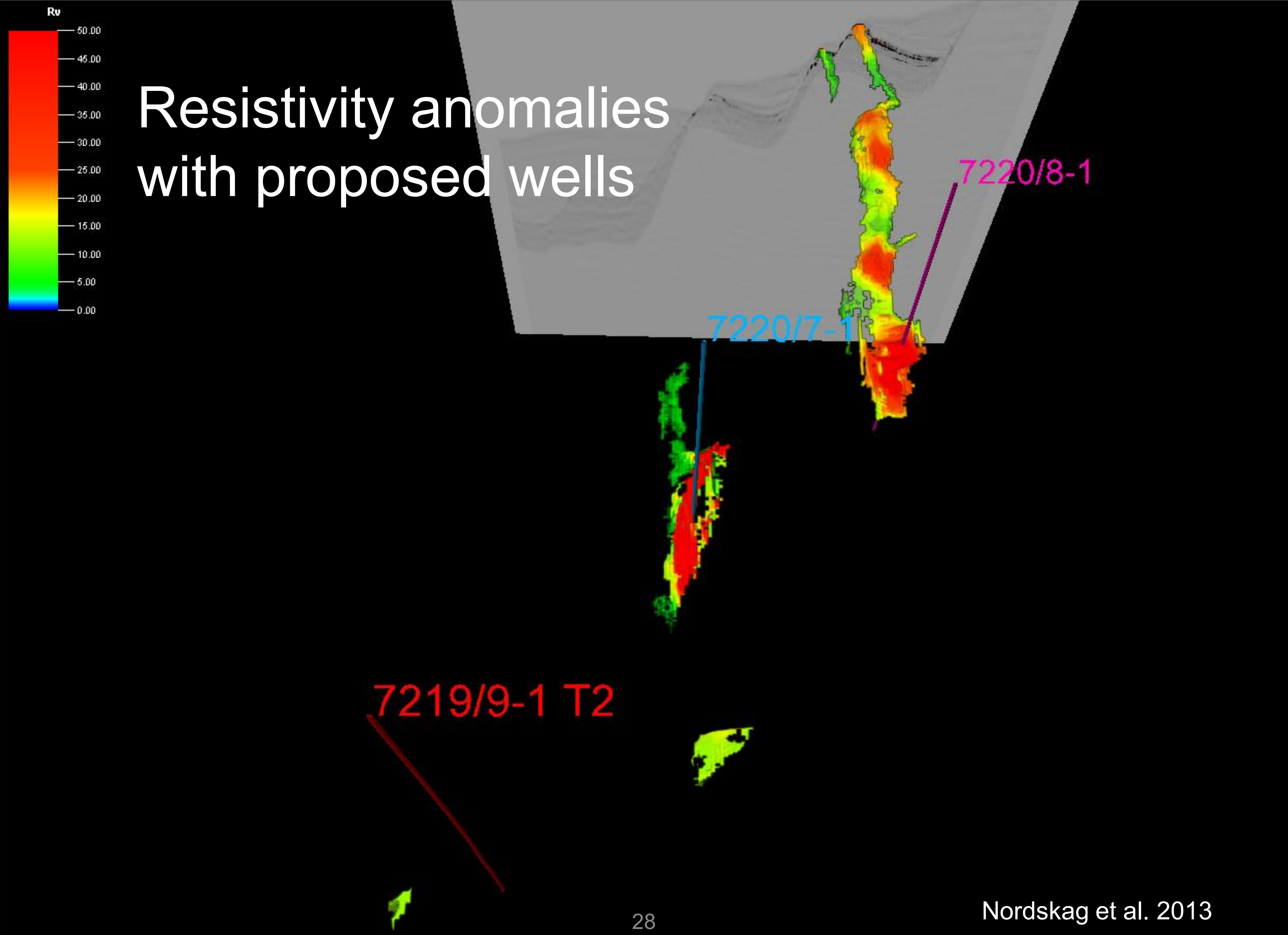
Gassmann vs. Archie equations

Structural conform seismic anomalies with proposed wells



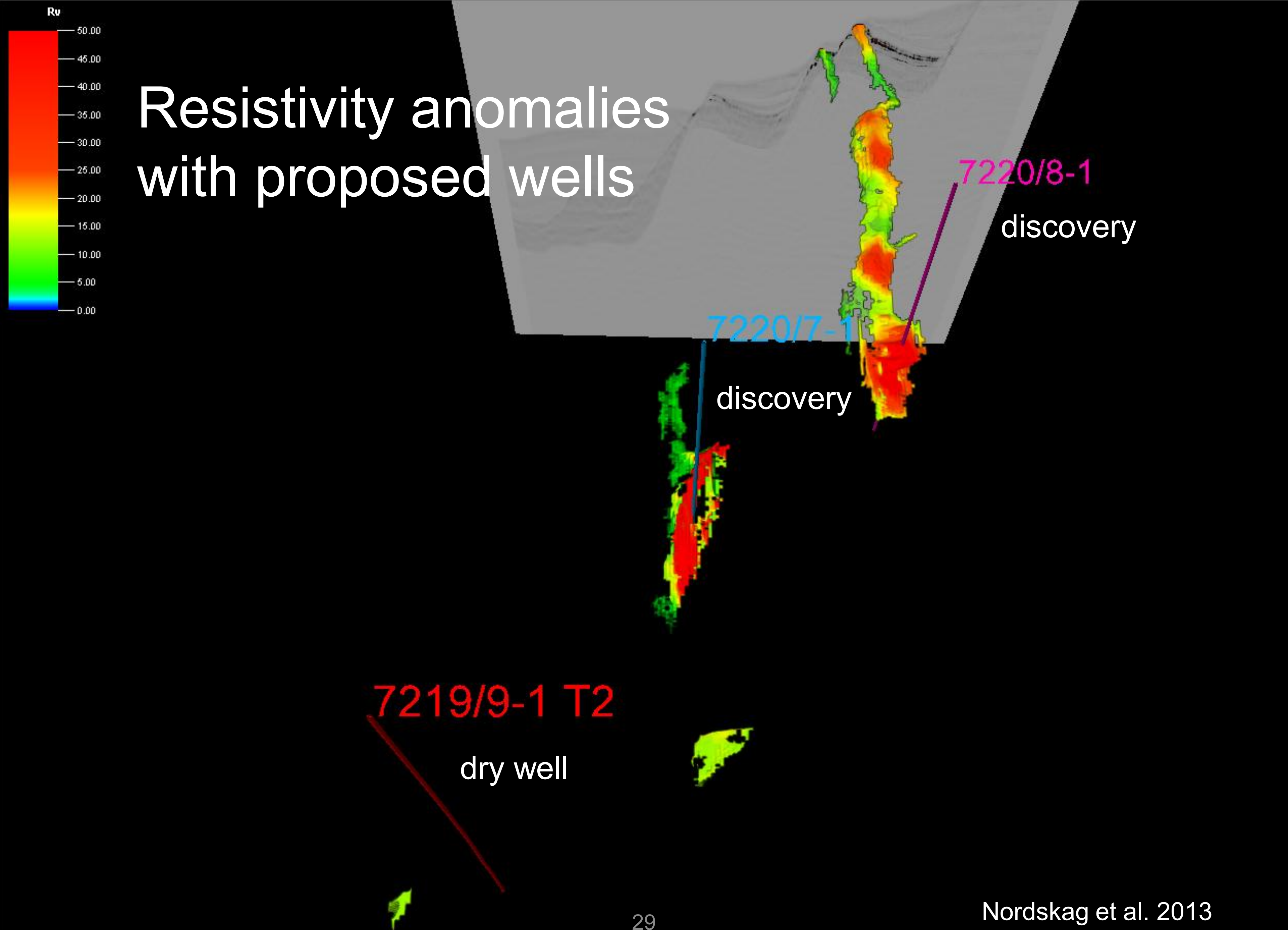
Nordskag et al. 2013

Resistivity anomalies with proposed wells



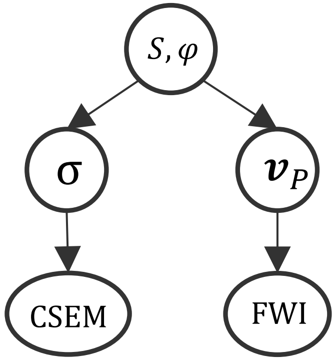
Nordskag et al. 2013

Resistivity anomalies with proposed wells



Nordskag et al. 2013

Fluid saturation and porosity from resistivity and seismic velocity



Rock physics forward models:

- Gassmann equation
$$v_p = f_v(S, \phi) + e_v$$
- Archie equation
$$R_t = f_R(S, \phi) + e_R$$
- Archie and Gassmann go into the likelihoods

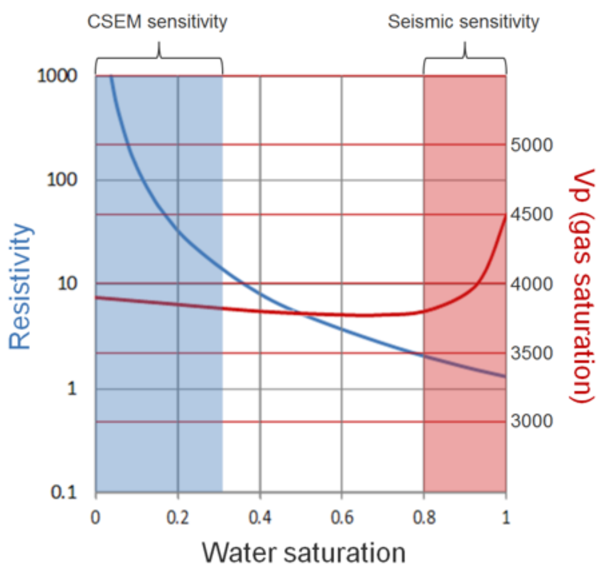
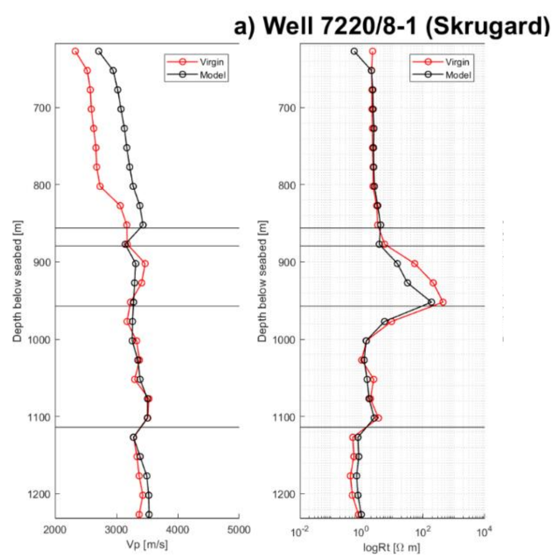


Figure from Constable, 2010



From MSc thesis of Sondre Olsson (2019)

Posterior distribution, mean and variance

Posterior distribution for saturation and porosity

$$p\left(S,\phi \middle| v_p, R_t;\lambda\right) \propto p\left(v_p \middle| S,\phi\right) p(R_t|S,\phi) p(S,\phi)$$

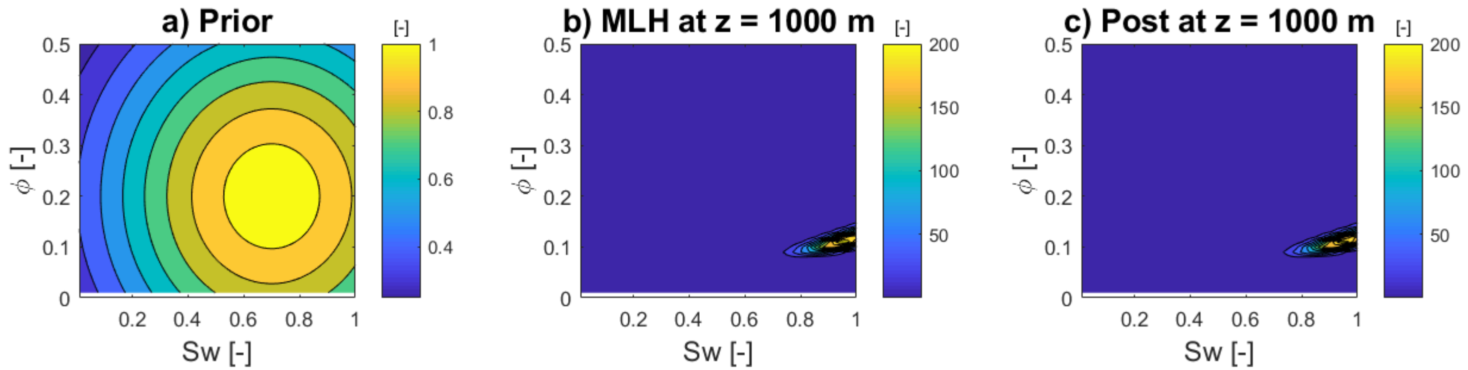
Substituting rock physics models in the likelihoods

$$p\left(S,\phi \middle| v_p, R_t;\lambda\right) \propto \underbrace{e^{-\frac{\left(v_p - f_v\right)^2}{2\sigma_{ev}^2}} e^{-\frac{(R_t - f_R)^2}{2\sigma_{eR}^2}}}_{\text{Likelihood}} \underbrace{e^{-\frac{\left(v_p - \mu_v\right)^2}{2\sigma_v^2}} e^{-\frac{(R_t - \mu_R)^2}{2\sigma_R^2}}}_{\text{Prior}}$$

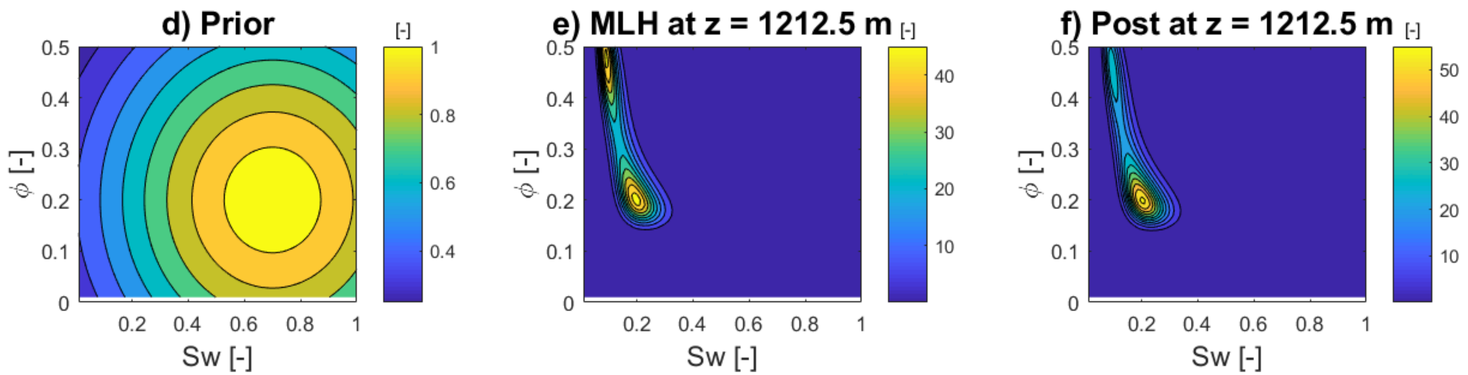
Posterior mean and variance computed from the definitions

Synthetic examples of distributions

Water case

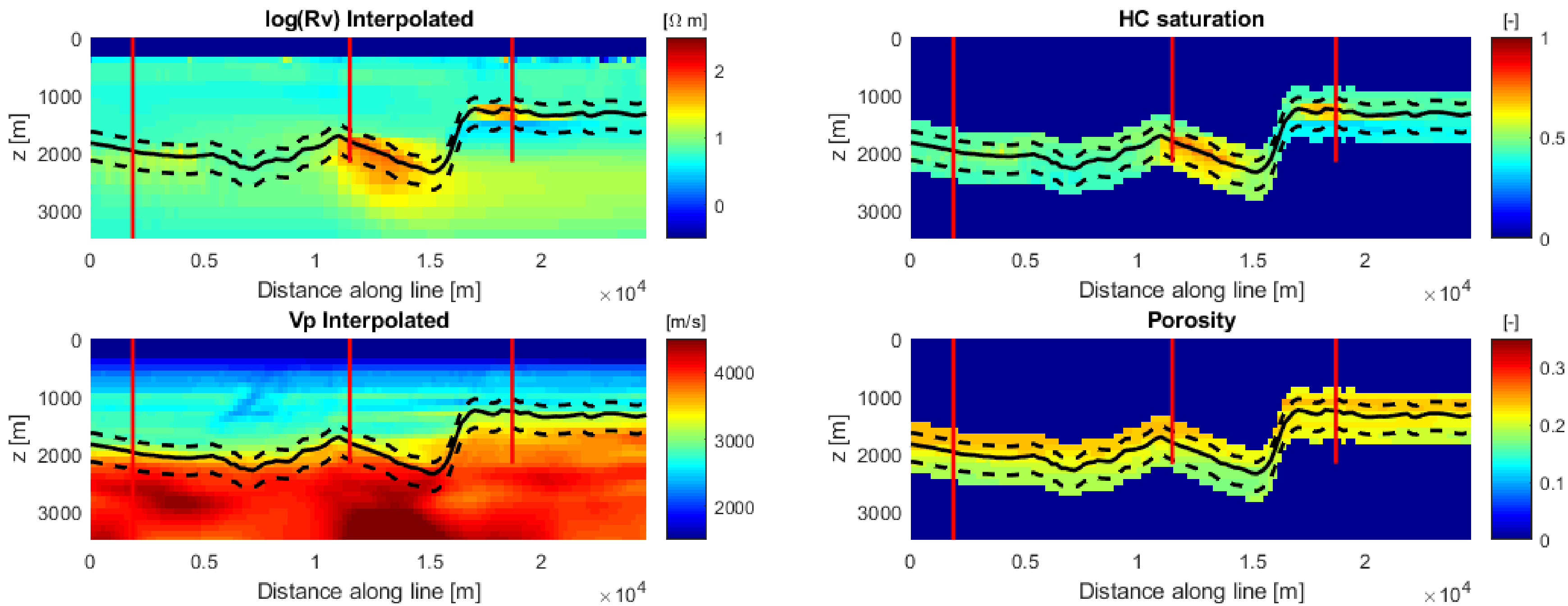


Hydrocarbon case



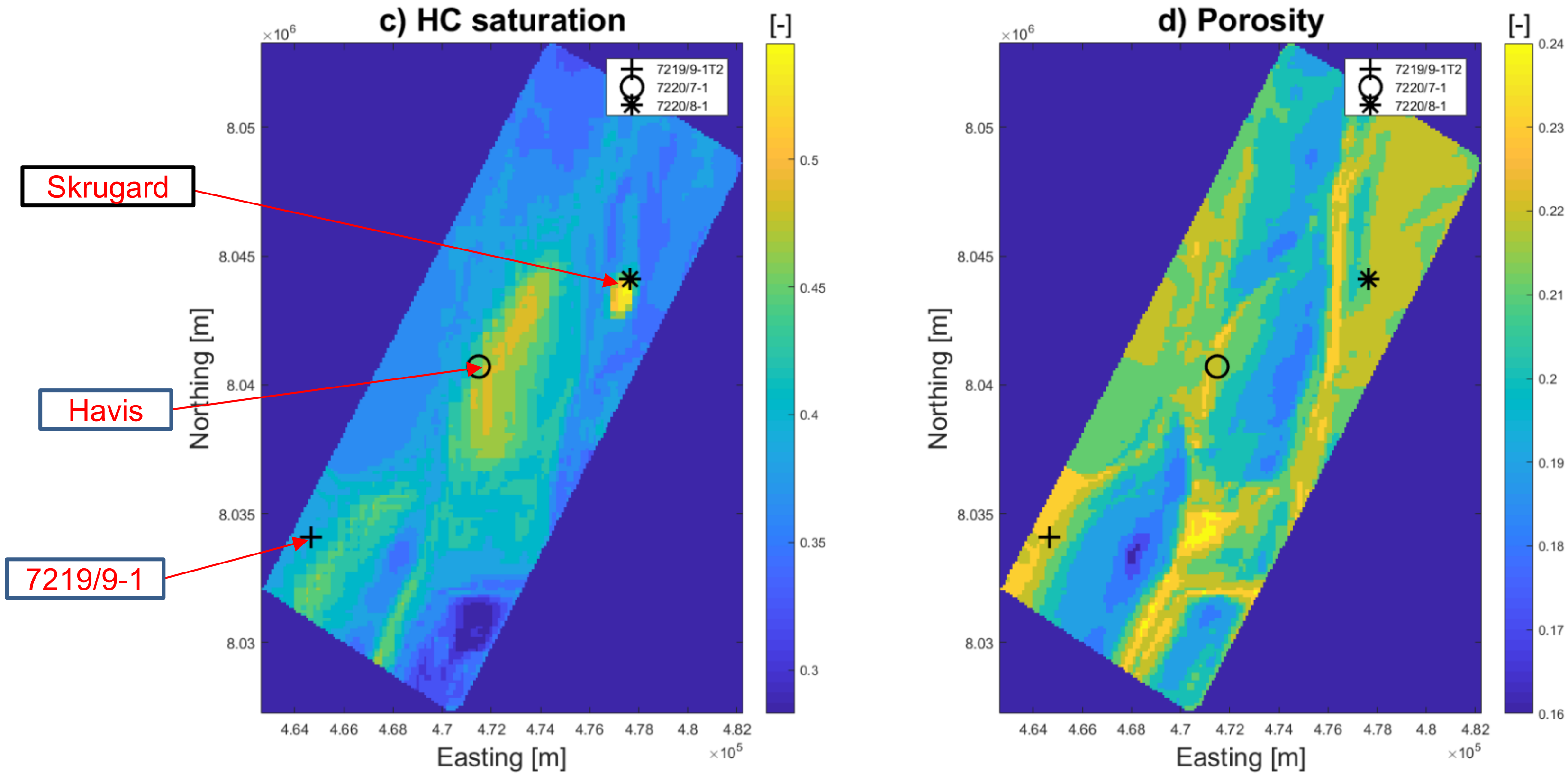
From MSc thesis of Sondre Olsson (2019)

Johan Castberg area – random line



From MSc thesis of Sondre Olsson (2019)

Johan Castberg area – map view
Realgrunnen Gp (Jurassic reservoir)



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